Practical No : 11

Tutorial Title: Write a C++ program to implement class and function templates.
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Class: SY **Div:** 2 **PRN No:** B25ET1099 **Batch:** B2

Program:

```
#include <iostream>
using namespace std;
template <typename T>
void displayResult(T result) {
   cout << result << endl; }
template <class T>
class Calculator {
private:
   T num1, num2;
public:
   Calculator(T n1, T n2) {
       num1 = n1;
       num2 = n2;  }
  T add() {
       return num1 + num2;   }
};
int main() {
   Calculator<int> c1(5, 4);
   cout << "The integer addition is: \t";
   displayResult(c1.add()); // Using the function template
 Calculator<double> c2(5.2, 4.3);
   cout << "The double addition is: \t";
   displayResult(c2.add());
  return 0;
}
```

Output:

main.cpp

Run ▶

```cpp
#include <iostream>
using namespace std;
template <typename T>
void displayResult(T result) {
    cout << result << endl; }
template <class T>
class Calculator {
private:
    T num1, num2;
public:
    Calculator(T n1, T n2) {
        num1 = n1;
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  T add() {
        return num1 + num2;   }
};
int main() {
    Calculator<int> c1(5, 4);
    cout << "The integer addition is: \t";
    displayResult(c1.add()); // Using the function template
 Calculator<double> c2(5.2, 4.3);
    cout << "The double addition is: \t";
    displayResult(c2.add());
 return 0;
```

Output

```
The integer addition is:    9
The double addition is:     9.5
```